

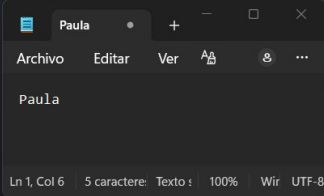
Ejercicio 1: nbtscan

Se utiliza nbtscan para descubrir equipos en la red local mediante consultas NetBIOS. Este comando envía peticiones a un rango de IPs y muestra los nombres NetBIOS, los servicios y las direcciones MAC.

```
(kali@kali)-[~]
└─$ sudo nbtscan -r 10.0.6.0/24
[sudo] password for kali:
Doing NBT name scan for addresses from 10.0.6.0/24
```

IP address	NetBIOS Name	Server	User	MAC address
10.0.6.11	<unknown>	<unknown>	<unknown>	
10.0.6.23	JOHNREDYSERV	<server>	JOHN REDY	08:00:27:99:79:09
10.0.6.255	Sendto failed: Permission denied			

```
(kali@kali)-[~]
└─$
```



Ejercicio 2.1: Detalles del sistema operativo con nmap

Se emplea el script smb-os-discovery de Nmap para obtener información del sistema operativo a través del puerto 445 (SMB).

`sudo nmap --script smb-os-discovery -p445 10.0.6.23`

```
(kali@kali)-[~]
└─$ sudo nmap --script smb-os-discovery -p445 10.0.6.23
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-07 06:12 EDT
Nmap scan report for 10.0.6.23
Host is up (0.00044s latency).
```

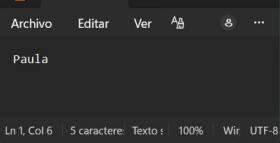
PORT	STATE	SERVICE
445/tcp	open	Microsoft-ds

MAC Address: 08:00:27:99:79:09 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:

```
| smb-os-discovery:
| OS: Windows 2000 (Windows 2000 LAN Manager)
| OS CPE: cpe:/o:microsoft:windows_2000::
| Computer name: johnredyserv
| NetBIOS computer name: JOHNREDYSERV\X00
| Workgroup: WORKGROUP\X00
|_ System time: 2026-05-07T13:12:02+02:00
```

Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds



Ejercicio 2.2: Versiones o dialectos de SMB soportados

Se utiliza el script smb-protocols para listar los dialectos SMB que soporta el servidor.

```
(kali@kali)-[~]
└─$ sudo nmap --script smb-protocols -p445 10.0.6.23
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-07 06:13 EDT
Nmap scan report for 10.0.6.23
Host is up (0.00054s latency).
```

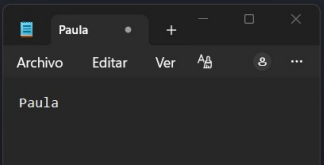
PORT	STATE	SERVICE
445/tcp	open	Microsoft-ds

MAC Address: 08:00:27:99:79:09 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:

```
| smb-protocols:
| dialects:
|_ NT LM 0.12 (SMBv1) [dangerous, but default]
```

Nmap done: 1 IP address (1 host up) scanned in 10.18 seconds



`sudo nmap --script smb-protocols -p445 10.0.6.23`

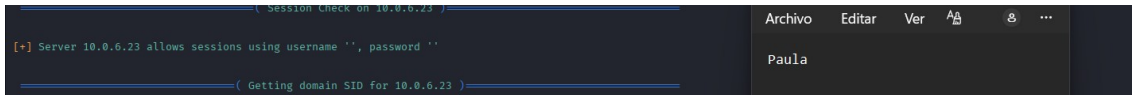
Ejercicio 3: enum4linux

enum4linux es una herramienta que automatiza la enumeración de recursos SMB (usuarios, shares, grupos, políticas, etc.).

`enum4linux 10.0.6.23`

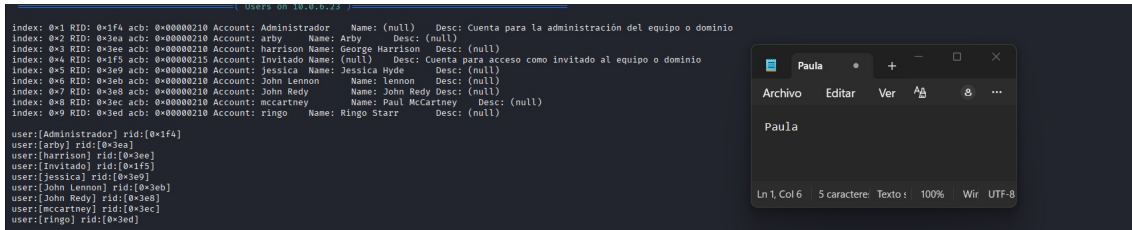
Ejercicio 3.1: Usuario y contraseña usados por enum4linux

Buscamos en la salida de enum4linux con qué credenciales se ha conectado la herramienta.



Ejercicio 3.3: Lista de usuarios y RID

Extraemos los usuarios del sistema y observamos sus RIDs.

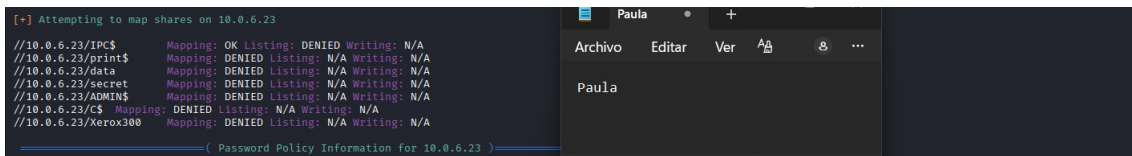


Los usuarios built-in (Administrador RID 0x1f4 = 500, Invitado 0x1f5 = 501) tienen RIDs bajos y fijos.

Los usuarios no built-in (arby, harrison, jessica, john redy, mccartney, ringo) tienen RIDs altos (entre 0x3e9 y 0x3ee), asignados dinámicamente. Además, son correlativos.

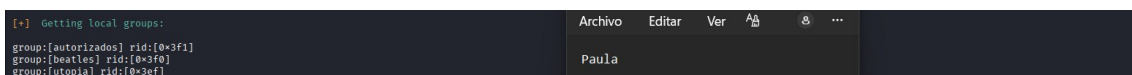
Ejercicio 3.4: Lista de shares

Enum4linux intenta listar los recursos compartidos.



Ejercicio 3.5: Grupos locales y sus miembros

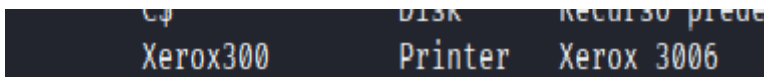
Enum4linux lista los grupos que no son built-in.



En el volcado de HE_SMB_EJ3_5.png no aparecen los miembros. Solo se muestran los grupos.

Sin embargo, más adelante en el ejercicio 4.1 se podrán obtener los miembros mediante otro script de Nmap.

Ejercicio 3.6: Impresora compartida



Ejercicio 4.1: Usuarios en grupos (script de nmap alternativo)

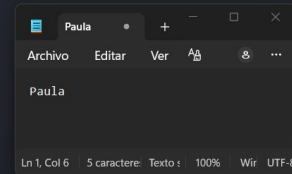
El script smb-enum-groups de Nmap permite enumerar grupos y sus miembros, información que enum4linux no mostró.

```
(kali@kali)~$ sudo nmap --script smb-enum-groups -p445 10.0.6.23
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-07 06:26 EDT
Nmap scan report for 10.0.6.23
Host is up (0.00055s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:99:79:09 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:
| smb-enum-groups:
|   BuiltinAdministradores (RID: 544): Administrador, John Redy
|   BuiltinUsuarios (RID: 545): Jessica, arby, John Lennon, mccartney, ringo, harrison
|   BuiltinInvitados (RID: 546): Invitado
|   BuiltinUsuarios avanzados (RID: 547): <empty>
|   BuiltinOperadores de copia (RID: 551): <empty>
|   BuiltinDuplicadores (RID: 552): <empty>
|   JOHNREDSERVutopia (RID: 1007): Jessica, arby
|   JOHNREDSERVbeatles (RID: 1008): ringo, mccartney, John Lennon, harrison
|   JOHNREDSERVautorizados (RID: 1009): John Redy, arby
|_

Nmap done: 1 IP address (1 host up) scanned in 0.32 seconds
```



Ejercicio 4.2: Vulnerabilidades

Se utiliza el script smb-vuln de Nmap para detectar vulnerabilidades conocidas en SMB.

```
sudo nmap --script smb-vuln -p445 10.0.6.23
```

```
(kali@kali)~$ sudo nmap --script smb-vuln* -p445 10.0.6.23
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-07 06:27 EDT
Nmap scan report for 10.0.6.23
Host is up (0.00035s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:99:79:09 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:
|_ smb-vuln-ms10-054: false
|_ smb-vuln-ms17-010:
|   VULNERABLE:
|     Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|     State: VULNERABLE
|     IDs: CVE:CVE-2017-0143
|     Risk factor: HIGH
|       A critical remote code execution vulnerability exists in Microsoft SMBv1
|       servers (ms17-010).
|
|     Disclosure date: 2017-03-14
|     References:
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
|       https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
|       https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt-attacks/
|_ smb-vuln-ms10-061: ERROR: Script execution failed (use -d to debug)
|_ smb-vuln-ms08-067:
|   VULNERABLE:
|     Microsoft Windows system vulnerable to remote code execution (MS08-067)
|     State: VULNERABLE
|     IDs: CVE:CVE-2008-4250
|       The Server service in Microsoft Windows 2000 SP4, XP SP2 and SP3, Server 2003 SP1 and SP2,
|       Vista Gold and SP1, Server 2008, and 7 Pre-Beta allows remote attackers to execute arbitrary
|       code via a crafted RPC request that triggers the overflow during path canonicalization.
|
|     Disclosure date: 2008-10-23
|     References:
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2008-4250
|       https://technet.microsoft.com/en-us/library/security/ms08-067.aspx
|_

Nmap done: 1 IP address (1 host up) scanned in 5.18 seconds
```

